

05 – Labrador Coast Hydro Screening Appendix

Project: Kristal Farms – Labrador Coast Partner Package

Document type: Level 2 due-diligence appendix

Status: Working draft for partner review

Primary source basis: *Potentiel hydroélectrique isolé au Nunavik et au Labrador (≥ 15 MW)*

Supporting source basis: *Kristal Farms – Cost Advantage & Strategic Rationale; Kristal Farms – Heat Recycling Plan; Kristal Farms Internal Reference Document; Northern Hydropower for Compute – Opportunities, Cooling Advantage & Strategy*

1. Purpose

This appendix screens hydroelectric opportunities relevant to the Kristal Farms Labrador coast thesis.

The goal is not to produce a final engineering study. The goal is to identify which hydro opportunities are aligned with the project model and which opportunities should be excluded before partner time is spent on them.

Kristal Farms is focused on **coastal-accessible, community-scale hydro / compute / heat opportunities**, beginning with **Nain** and then replicating along the Labrador coast where the same conditions exist.

The project is **not** pursuing inland megaprojects, long-distance transmission plays, or large dam development as the first route to market.

2. Executive Summary

The current source base supports one clear priority hydro candidate:

Priority	Site / community	Hydro basis	Current status	Treatment in package
1	Nain / Fraser River	Preliminary 15–20 MW run-of-river/community hydro concept	Highest-priority first target	Carry forward as the first Labrador coast target, subject to validation
2	Hopedale	No specific river candidate validated in current source package	Replication community	Carry forward as a replication-screening community, not as a confirmed site
3	Makkovik	No specific river candidate validated in current source package	Replication community	Carry forward as a replication-screening community, not as a confirmed site
4	Postville	No specific river candidate validated in current source package	Replication community	Carry forward as a replication-screening community, not as a confirmed site
5	Rigolet	No specific river candidate validated in current source package	Replication community	Carry forward as a replication-screening community, not as a confirmed site

The priority conclusion is:

Nain is the first target because it is the only Labrador coast site in the current source package with a specific hydro candidate, a plausible community-scale capacity range, port access, and alignment with the village-first compute/heat model.

The appendix also records why certain opportunities should **not** be treated as first-wave targets:

- large inland or transmission-oriented projects;
 - operating or already-planned hydro assets;
 - projects that require long HV corridors;
 - Nunavik mega-river concepts with major environmental, logistical, or political complexity;
 - sites with no validated local heat sink, no fibre path, or no feasible coastal logistics plan.
-

3. Screening Thesis

Kristal Farms should screen hydro opportunities using the following thesis:

Use local coastal hydro to power village-sited compute pads, recycle server heat locally, and export compute outputs by fibre instead of exporting electricity over long transmission lines.

This leads to four practical rules.

Rule 1 — The compute pad should be near heat users, not at the dam

The current heat architecture is village-first. Containers should sit near the village, port, public buildings, housing clusters, and greenhouse heat sinks. A short medium-voltage feed can connect the hydro source to the village substation.

A site is weaker if it requires containers to sit far from heat users, because the main community value is lost.

Rule 2 — Export compute, not electrons

The model avoids long high-voltage transmission corridors. The hydro source powers the local pad yard and community load. Results move by fibre.

A site is weaker if the main value depends on exporting power over long transmission lines.

Rule 3 — Heat value matters as much as hydro capacity

Hydro potential alone is not enough. A site should have nearby useful heat demand:

- clinic;
- school;
- municipal buildings;
- housing clusters;
- domestic hot-water preheat;
- greenhouse heat demand;
- thermal storage potential.

A site with strong hydro but no heat sink is less aligned with the Kristal Farms partner package.

Rule 4 — The first project should be modular and reversible

The first site should allow staged deployment:

1. site validation;
2. community/FPIC process;
3. hydro and grid pre-feasibility;
4. fibre and logistics design;
5. pilot pad;
6. heat loop and greenhouse;
7. expansion pads.

Sites that require one large irreversible capital decision before proof of value are less suitable for the first wave.

4. Screening Criteria

Each hydro candidate should be scored against the following criteria.

Criterion	Question	Strong signal	Weak signal
Hydro resource	Is there enough year-round flow and head?	Preliminary 15–20 MW+ potential with winter-flow path to validation	Only theoretical annual flow; no winter-flow data
Seasonal reliability	Can the site support winter operation?	Documented under-ice flow, frazil management plan, seasonal flow curve	High spring/summer flow only; winter uncertainty
Coastal access	Can heavy equipment arrive by sea?	Port, sheltered landing, crane plan, short road route	Long inland road, difficult overland transport
Distance to village	Can power reach a village substation by short MV line?	Short MV route, no long HV corridor	Long transmission line needed
Heat users	Are heat sinks close to the pad yard?	Public buildings, housing, greenhouse site, short heat loop	No concentrated heat demand
Fibre path	Can compute outputs be exported reliably?	Existing/planned fibre, redundancy path, NOC concept	No credible fibre path
Environmental sensitivity	Can impacts be minimized?	Run-of-river, small/no reservoir, non-contact cooling, limited inundation	Large reservoir, protected habitat, fish/marine conflicts

Criterion	Question	Strong signal	Weak signal
Community consent	Can FPIC and benefit-sharing proceed credibly?	Clear governance pathway and local benefit case	Unclear consent structure or unresolved land issues
Constructability	Can the work fit northern construction windows?	Seasonal marine logistics plan, modular components	Complex civil works, long exposed schedule
Replicability	Does the site teach a repeatable model?	Coastal village + hydro + port + heat + fibre pattern	One-off special case

5. Current Candidate Inventory

5.1 Priority Candidate — Nain / Fraser River

Field	Current source position	Partner-package treatment
Community	Nain, Nunatsiavut, Labrador coast	First target
Candidate hydro source	Fraser River	Carry forward as preliminary candidate
Indicative capacity	15–20 MW	Treat as preliminary; validate before partner financial claims
Configuration	Run-of-river/community hydro concept	Use as likely model, subject to engineering validation
Distance / integration	Source describes concept near Nain with village and data-center integration	Validate route, grade, land access, interconnection, and heat users

Field	Current source position	Partner-package treatment
Logistics	Marine delivery through Nain port, with possible port upgrades	Validate sealift, crane, quay capacity, staging, and road access
Existing diesel context	Nain currently relies on diesel generation in the source narrative	Use as strategic rationale; verify current installed capacity and fuel volumes
Compute fit	Modular data center / AI compute is explicitly linked in the source narrative	Carry forward as core first-site rationale
Status	Planning / concept level in the source package	Not yet bankable; needs hydrology, geotech, EIA, FPIC, fibre, and cost validation

Why Nain is first

Nain is the correct first target because it aligns with all major Kristal Farms filters:

1. **Labrador coast focus** — it is a coastal Labrador community, not an inland megaproject.
2. **Specific hydro candidate** — the current source package identifies the Fraser River candidate.
3. **Community-scale range** — the preliminary 15–20 MW range fits modular compute plus local load better than a mega-dam frame.
4. **Port logic** — equipment can be planned around marine logistics.
5. **Heat-first model** — the village contains the heat users; containers should be placed near public buildings, housing, and greenhouse users.
6. **Replication logic** — if the pattern works in Nain, other Labrador coast communities can be screened using the same template.

What must be validated before partner commitment

The following items must be validated before Nain is presented as investment-ready:

- monthly and seasonal flow profile;
- winter low-flow and under-ice flow availability;

- frazil ice and intake-freezing risks;
- usable head and penstock route;
- run-of-river design feasibility;
- reservoir/pondage footprint, if any;
- fish passage and aquatic habitat impacts;
- geotechnical conditions at intake, powerhouse, road, and substation sites;
- route and cost of MV connection to the village substation;
- port landing and heavy-lift capacity;
- road-access route from port/village to hydro works;
- fibre availability and redundancy plan;
- local heat-load inventory;
- FPIC and Nunatsiavut governance pathway;
- updated cost estimate with northern contingency.

5.2 Replication Communities — Screening Inventory

The following communities are part of the Labrador coastal replication logic, but the current source package does not yet validate specific hydro sites for them.

Community	Role in package	Current hydro status	Required next step
Hopedale	Replication community	No confirmed candidate in current source package	Desktop hydro scan: nearby rivers, head, flow, access, protected areas
Makkovik	Replication community	No confirmed candidate in current source package	Desktop hydro scan and heat-load inventory
Postville	Replication community	No confirmed candidate in current source package	Desktop hydro scan, port/logistics check, local load baseline

Community	Role in package	Current hydro status	Required next step
Rigolet	Replication community	No confirmed candidate in current source package	Desktop hydro scan with special attention to coastal access and environmental constraints

These communities should not be described as confirmed hydro candidates. They should be described as **replication-screening communities**.

The correct partner-facing language is:

“Nain is the first target. Hopedale, Makkovik, Postville, and Rigolet are replication-screening communities where the same coastal hydro / compute / heat logic can be tested after site data is gathered.”

6. Excluded or Deferred Site Categories

6.1 Inland megaprojects

Large inland hydropower assets and large transmission-oriented opportunities should be excluded from the first partner package.

Examples include:

- Churchill Falls;
- Muskrat Falls;
- Gull Island or other large-scale Labrador hydro concepts.

These assets may be strategically important in a broad Canadian energy discussion, but they are not the Kristal Farms first-wave model.

Reasons for exclusion:

- they are not village-scale coastal replication opportunities;
- they imply large institutional power-sector structures;
- they can distract from the Nain-first thesis;
- they generally do not solve the heat-reuse problem for coastal villages;
- they can pull the conversation toward grid export rather than compute export.

Partner-facing rule:

Kristal Farms is not pursuing mega-dams or inland transmission projects as its first route to market.

6.2 Existing operating or already-planned hydro assets

Operating or already-committed hydro projects may be relevant as comparables, but should not be treated as Kristal Farms target sites unless there is a clear new pad, heat, and fibre opportunity.

Examples from the broader source material include:

- Innavik / Inukjuak as a useful remote-hydro precedent;
- Iqaluit / Kuugaluk as a useful Arctic hydro planning precedent.

These are **case studies**, not Labrador coast target sites.

6.3 Nunavik mega-river concepts

The broader hydro source includes large Nunavik rivers with major theoretical potential. These are not first-wave Kristal Farms targets.

Examples include:

- Grande Rivière de la Baleine;
- Nastapoka;
- George River;
- Rivière aux Feuilles;
- Arnaud;
- Koksoak;

- Rivière à la Baleine.

Reasons for exclusion or deferral:

- not Labrador coast;
- major environmental sensitivity;
- large reservoir or diversion risk;
- permafrost and methane risk in some large-reservoir scenarios;
- heavy social and consent history;
- very high logistics burden;
- weak fit with the “first coastal Labrador village” package.

Nunavik material can remain in the source archive and may be used only as comparison or exclusion logic.

7. Hydro Evaluation Method

The source hydro methodology is based on a practical hydro-screening logic: flow, head, seasonality, winter operability, access, environmental acceptability, and social legitimacy.

7.1 Hydrologic resource

The first technical question is whether the river has enough usable flow.

Required data:

- watershed area;
- precipitation and snowmelt pattern;
- monthly flow curve;
- annual average flow;
- winter low-flow estimate;
- flood-frequency data;
- climate-change sensitivity;
- uncertainty range.

The key partner question is not “what is the theoretical annual capacity?”

The key partner question is:

How much reliable power is available when the village and compute campus need it, especially in winter?

7.2 Head and site geometry

The second technical question is whether the river has enough exploitable head without creating a large reservoir.

Required data:

- topographic profile;
- LiDAR or photogrammetry;
- rapids / falls / constriction inventory;
- possible intake location;
- possible powerhouse location;
- penstock route;
- road route;
- geotechnical and bedrock conditions.

A strong Kristal Farms site should have a useful head/flow combination that supports a **run-of-river or low-pondage design**.

A weak site requires large inundation, long canals, or major civil works before a modest amount of dependable power can be produced.

7.3 Power estimate

For early screening, the power estimate can be approximated from:

$$P = \rho \times g \times Q \times H \times \eta$$

Where:

- P = power output;
- ρ = water density;

- g = gravitational acceleration;
- Q = flow;
- H = head;
- η = efficiency.

For practical pre-screening, the source document uses the rule of thumb:

$$P(\text{kW}) \approx 7 \text{ to } 8 \times Q(\text{m}^3/\text{s}) \times H(\text{m})$$

This should be treated as a **screening estimate only**, not an investment-grade capacity claim.

7.4 Winter operability

Northern sites require a dedicated winter-operability screen.

Required checks:

- river ice cover;
- anchor ice;
- frazil ice;
- intake clogging;
- minimum flow under ice;
- access to intake in winter;
- line and substation reliability;
- emergency diesel/battery backup;
- seasonal maintenance windows.

A site that looks strong on annual average flow can fail if winter low flow is too weak or if intake icing cannot be managed.

7.5 Environmental acceptability

Environmental screening should be completed before any partner-facing claim that a site is “low impact.”

Required checks:

- fish habitat and fish passage;
- salmonid / Arctic char / migratory species issues;
- wetlands;
- marine outlet impacts;
- protected areas;
- bird and terrestrial habitat;
- archaeological/cultural sites;
- permafrost and terrain disturbance;
- mercury/methylmercury risk if pondage is created;
- cumulative effects;
- construction sediment controls;
- environmental flow requirements.

A site should be preferred if it can use a low-impact design with minimal inundation and a clear environmental flow plan.

7.6 Coastal logistics

Coastal access is a core criterion, not a minor construction detail.

Required checks:

- sealift window;
- berth and draft limits;
- crane/heavy-lift capacity;
- temporary landing possibility;
- staging area near port;
- route from port to hydro works;
- bridge and culvert needs;
- fuel, camp, and equipment storage;
- weather delay contingency;
- demobilization and reversibility.

A site should be downgraded if heavy equipment requires a long new road, repeated overland hauling, or construction outside reliable seasonal windows.

7.7 Electrical integration

The preferred project architecture is a short MV connection from hydro to a village substation.

Required checks:

- route length;
- voltage level;
- right-of-way;
- substation location;
- power quality;
- protection coordination;
- pad metering;
- reserve margin for community load;
- emergency backup arrangements;
- expansion capacity.

Sites requiring long HV lines are not aligned with the first package.

7.8 Compute, fibre, and heat fit

Hydro screening must be connected to the compute and heat model.

Required checks:

- pad yard location near port or village edge;
- fibre route and redundancy;
- NOC location;
- public-building heat demand;
- housing heat demand;
- greenhouse site;
- thermal storage area;
- cold-source access at bay/port;
- non-contact heat rejection feasibility;
- winter and summer heat-sink balance;
- public dashboard and metering plan.

A hydro site should not be ranked highly unless it can support **power + fibre + heat reuse + community governance** together.

8. Site Ranking Logic

The following scoring system should be used for future candidate screening.

Score	Meaning
5	Strong fit; clear positive evidence; ready for pre-feasibility work
4	Good fit; some data gaps but likely solvable
3	Possible fit; major unknowns remain
2	Weak fit; one or more core constraints likely problematic
1	Poor fit; not aligned with project thesis
0	Excluded

Recommended weightings

Criterion	Weight
Hydro resource and winter reliability	20%
Coastal logistics and access	15%
Distance to village / MV practicality	15%
Heat-use potential	15%
Fibre and compute operations	10%
Environmental acceptability	10%
Community/FPIC pathway	10%

Criterion	Weight
Replicability	5%

Current preliminary ranking

Site / category	Current score	Rationale
Nain / Fraser River	4	Strongest current fit; specific candidate and project concept exist; still requires validation
Hopedale	3	Replication community, but no validated hydro candidate in current source package
Makkovik	3	Replication community, but no validated hydro candidate in current source package
Postville	3	Replication community, but no validated hydro candidate in current source package
Rigolet	3	Replication community, but no validated hydro candidate in current source package
Churchill Falls / Muskrat Falls / Gull Island category	0	Excluded from first package as inland/megaproject/transmission-oriented category
Nunavik mega-river category	0-1	Useful comparison/archive material, not first Labrador coast package

These are **screening scores**, not engineering rankings.

9. Required Next Studies

9.1 Nain / Fraser River — immediate diligence package

The Nain diligence package should be organized as follows.

A. Hydrology

- install or obtain stream-gauge data;
- build monthly flow model;
- calculate winter firm flow;
- estimate annual energy production;
- model climate sensitivity;
- define environmental flow requirements.

B. Topography and geotechnical

- LiDAR survey;
- intake/powerhouse reconnaissance;
- penstock route survey;
- road-access alignment;
- bedrock and foundation investigation;
- permafrost and slope-stability screening.

C. Environmental baseline

- fish habitat;
- spawning areas;
- marine outlet conditions;
- wetlands and riparian zones;
- wildlife and bird use;
- archaeological and cultural sites;
- mercury/methylmercury risk if pondage is used;
- construction sediment control.

D. Community and FPIC

- identify decision-making bodies;
- define FPIC process timeline;

- define information-sharing process;
- define benefit-sharing principles;
- define grievance process;
- define local employment/training pathway;
- define heat allocation priorities.

E. Port and logistics

- confirm quay/landing capacity;
- confirm crane/heavy-lift needs;
- map staging area;
- define sealift schedule;
- define temporary road needs;
- price mobilization and demobilization;
- create weather-delay contingency.

F. Electrical integration

- select MV voltage concept;
- define hydro-to-village feeder route;
- identify village substation site;
- define pad-level metering;
- define protection coordination;
- define emergency backup and black-start requirements.

G. Fibre and compute operations

- confirm southbound fibre path;
- assess redundancy and outage risk;
- define NOC requirement;
- define pad bandwidth assumptions;
- define tenant black-box boundary;
- define service monitoring fields.

H. Heat and cold-source integration

- inventory public-building heating systems;
- measure current fuel use and heat demand;
- identify first heat customers: clinic, school, municipal buildings;
- assess housing cluster extensions;
- identify greenhouse location;
- confirm bay/seawater non-contact cooling concept;
- size first thermal storage tank;
- define HUF, MWh-th, PUE, WUE, diesel-avoided metrics.

9.2 Replication communities — desktop screening package

For Hopedale, Makkovik, Postville, and Rigolet, the first study should be a desktop screen, not a field campaign.

For each community, collect:

- nearby river candidates;
- watershed size;
- preliminary head from DEM;
- approximate river-to-village distance;
- port or landing condition;
- protected area constraints;
- public-building heat demand;
- diesel generation baseline;
- fibre status;
- possible pad-yard location;
- possible greenhouse location;
- local governance pathway;
- fatal-flaw risks.

The output should be a one-page screening sheet per community.

10. Evidence Standards

The partner package should use three evidence labels.

Confirmed

Use only when supported by verified documents, public records, engineering studies, or signed partner/community information.

Preliminary

Use for claims supported by source material but not yet independently validated.

Example:

“The Fraser River candidate near Nain is treated as a preliminary 15–20 MW opportunity based on the current source package, subject to hydrology, engineering, environmental, cost, and community validation.”

Open / To validate

Use for all replication communities until a specific candidate river and site geometry are identified.

Example:

“Hopedale is included as a replication-screening community. A specific hydro candidate has not yet been validated in the current source package.”

11. Partner-Facing Language to Use

Use this language in Level 1 documents:

“Kristal Farms begins with Nain because the current source base identifies a plausible community-scale hydro candidate near the community, and because Nain fits the coastal-access, heat-reuse, fibre-export, modular-pad model. Additional Labrador coast communities will be screened using the same method after the Nain diligence pathway is defined.”

“The project is not a mega-dam or long-transmission proposal. It is a coastal community infrastructure model: local hydro, village-sited compute, useful heat, fibre export, and phased modular deployment.”

“All hydro capacity figures are preliminary until validated through hydrology, winter operability, geotechnical, environmental, cost, and community-consent studies.”

12. Language to Avoid

Avoid the following in partner-facing materials:

- “Hydro site confirmed” unless engineering validation is complete.
 - “Guaranteed 15–20 MW” for Nain.
 - “Year-round marine access” unless specifically validated.
 - “No environmental impact.”
 - “Community consent secured” unless formally true.
 - “Nain model applies automatically to all Labrador communities.”
 - “Churchill Falls / Muskrat Falls / Gull Island are target sites.”
 - “Nunavik mega-river opportunities are part of the first package.”
 - Any unverified cost, revenue, IRR, tariff, or PUE claim.
-

13. Diligence Questions for Partners

Hydro / utility partner

1. What hydrology data exists for the Fraser River?
2. What winter low-flow assumptions are defensible?
3. What interconnection voltage is appropriate for a village-scale pad yard?
4. What reserve margin must remain for community loads?
5. What backup standard is required for critical loads?

Infrastructure investor

1. What evidence is required before funding pre-feasibility?
2. What cost contingency is appropriate for Labrador coastal construction?
3. What decision gates are needed before construction commitment?
4. What assets remain useful if compute demand changes?
5. What parts of the project are reversible or modular?

Community / government partner

1. What FPIC process applies?
2. Who must be consulted and in what order?
3. What benefit-sharing model is acceptable?
4. What local heat priorities should be written into the project charter?
5. What jobs and training commitments should be binding?

Telecom / fibre partner

1. What fibre route serves Nain?
2. What redundancy is possible?

3. What outage history exists?
4. What bandwidth is realistic for first pad and expansion pads?
5. What service-level commitments can be supported?

Compute tenant

1. What power envelope can the first pad support?
2. What cooling interface is available?
3. What uptime and fibre SLA is realistic?
4. What telemetry does the host see?
5. How is tenant data excluded from host visibility?

14. Appendices to Add Later

The following attachments should be added when available:

maps/

kristal_farms_LABRADOR_COAST_PROJECT_ONLY_colored.kml

kristal_farms_LABRADOR_COAST_PROJECT_ONLY_feature_inventory.csv

kristal_farms_LABRADOR_COAST_PROJECT_ONLY_validation.csv

site_sheets/

Nain_Fraser_River_site_sheet.md

Hopedale_screening_sheet.md

Makkovik_screening_sheet.md

Postville_screening_sheet.md

Rigolet_screening_sheet.md

technical_inputs/

Nain_hydrology_data_request.md

Nain_port_logistics_data_request.md

Nain_heat_load_inventory_template.md

Labrador_coast_replication_screening_template.md

15. Source Notes

This appendix is based primarily on the uploaded hydro screening source, which identifies broad Nunavik/Labrador hydro potential and includes the Nain/Fraser River candidate. It is adapted to the current Kristal Farms package architecture by filtering out material that does not serve the Labrador coast, Nain-first, village-heat, modular-compute thesis.

The heat-first and village-first rules come from the uploaded Heat Recycling Plan and Internal Reference Document. The cost-screening rules come from the Cost Advantage & Strategic Rationale source. The large northern hydro context, including Churchill Falls / Muskrat Falls / Gull Island as broad northern capacity examples, is treated only as background and exclusion logic for this appendix.

16. Bottom Line

The hydro screening conclusion is simple:

Carry Nain / Fraser River forward as the first Labrador coast hydro candidate, but only as a preliminary candidate until site data is validated. Treat Hopedale, Makkovik, Postville, and Rigolet as replication-screening communities. Exclude inland megaprojects, Nunavik mega-river concepts, and long-transmission opportunities from the first partner package.